

Rishon Ghosh

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Education

Simon Fraser University

Bachelor of Science, Computer Science (Joint Major with Business) | **GPA: 3.9**

Burnaby, BC, Canada

Expected Grad Date: Dec 2027

- **Coursework:** Algorithms & Data Structures, Introduction to Software Engineering, Business Technology Management
- **Awards:** Dean's Honour Roll, President's Honour Roll, SFU Undergraduate Open Scholarship

Experience

Software Engineer Consultant

Self-Employed

Vancouver, BC

Dec 2025 - Present

- Architected Python and Flask REST backends for [X] small-business clients, turning existing sales and inventory data into reports
- Engineered SQL schemas and ETL scripts consolidating spreadsheet and POS data, saving [X] **hours** of manual entry weekly
- Optimized database performance through indexing and query refactoring, reducing average API response times by [X]%
- Implemented secure authentication with password hashing and input validation to protect client and customer records
- Executed client discovery, scoping, and sprint planning to translate business needs into technical requirements delivered on schedule

Projects

FlashCart | *Go, TypeScript, React, AWS Lambda, API Gateway, DynamoDB, SQS, CDK*

Sep 2026

- Spearheaded a serverless flash-sale API in Go on AWS Lambda and API Gateway, load-tested with k6 at **1,000+** concurrent users
- Devised oversell-proof inventory reservations with DynamoDB conditional writes and idempotency keys, recording **0** oversold units
- Orchestrated asynchronous order fulfillment through SQS with dead-letter queue retries, holding checkout p99 latency under **150ms**
- Established infrastructure as code with AWS CDK in TypeScript and GitHub Actions CI/CD, deploying the full stack on every merge
- Launched a React and TypeScript dashboard on S3 and CloudFront that visualizes live stock levels and CloudWatch latency metrics

MatchMesh | *C++20, gRPC, Protocol Buffers, TypeScript, Angular, Docker*

Sep 2026

- Pioneered a fault-tolerant distributed matchmaking cluster in C++20, sharding player queues across **5** nodes with consistent hashing
- Directed inter-node communication over gRPC and Protocol Buffers, forming skill-balanced lobbies in under **50ms** at the median
- Championed resilience with heartbeat failure detection and primary-backup replication, recovering failed shards within **2s**
- Drove thread-safe queueing with a worker pool and lock striping, stress-testing **10,000** simulated players with ThreadSanitizer
- Instituted chaos testing with Docker Compose and an Angular dashboard that streams live cluster topology over WebSockets

Graphite Gambit | *Java, libGDX, JUnit, Mockito, Maven, JaCoCo*

Jan 2026

- Led a team building a 2D top-down game in Java and libGDX, establishing weekly agile sprints and a merge-conflict protocol
- Created a map-loading system and collision engine, using mathematical proofs in testing to verify safe entity movement and bounds
- Automated a testing suite with JUnit and Mockito, achieving **71%** branch coverage across core game packages

Rubik's Cube Solver | *Java*

Oct 2025

- Programmed an optimal Rubik's Cube solver in Java using Iterative Deepening A* (IDA*) search, finding solutions in under **10s**
- Constructed memory-efficient pattern databases using combinatorial indexing and factorial mapping for $O(1)$ heuristic lookups
- Compressed a **43-quintillion** state space into a **~2MB** footprint, keeping heuristic data fully in memory during search

Technical Skills

Languages: Go, C++, TypeScript, Java, Python, SQL, C, JavaScript, HTML/CSS

Frameworks & Libraries: React, Angular, gRPC, Protocol Buffers, Flask, libGDX, JUnit, Mockito, Google Test

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, SQS, S3, CloudFront, CloudWatch, CDK), Docker, GitHub Actions

Databases: DynamoDB, SQLite

Developer Tools: Git, CMake, Maven, k6, JaCoCo, ThreadSanitizer, VS Code

Concepts: Distributed Systems, Fault Tolerance, Consistent Hashing, Concurrency, REST APIs, CI/CD, Infrastructure as Code

Business: Financial Accounting, Excel, PowerPoint, Sprint Planning & Execution